Prob 1 (a) use SES update formula: $l_{10} = \alpha y_{10} + (1-\alpha) l_9$

$l_{10} = 0.3(280) + 0.7(250) = 259$ cups

(b) $\hat{y}_{10|9} = l_9 = 250$ cups

$e = y_{10} - \hat{y}_{10|9} = 280 - 250 = 30$ cups

(c) $\hat{y}_{11|10} = l_{10} = 259$ cups.

(d) $l_t = \alpha y_t + (1-\alpha) l_{t-1}$

rearranging: $l_t = \alpha y_t + l_{t-1} - \alpha l_{t-1}$

$= l_{t-1} + \alpha (y_t - l_{t-1})$.

where $y_t - l_{t-1}$ is the one-step forecast error.

(e) when $\alpha = 0.7$, $l_{10} = 0.7(280) + 0.3(250) = 271$ cups

The new level estimates would be higher.

from (d) we know the FOC w.r.t. α of l_t is the forecast error $(y_t - l_{t-1})$

as long as $y_t - l_{t-1} > 0$, $\frac{\partial l_t}{\partial \alpha} > 0$, ie. $\alpha \uparrow$, $l_t \uparrow$

Prob 2 (a) $l_6 = \alpha y_6 + (1-\alpha)(l_{5} + b_{5}) = 10,780$ subscribers

(b) $b_t = \beta(l_t - l_{t-1}) + (1-\beta) b_{t-1}$

$b_6 = 556$ subscribers/month.

(c) $\hat{y}_{7|6} = l_6 + b_6 = 11336$ $\hat{y}_{8|6} = l_6 + 2b_6 = 11892$

Prob 3 (a) $MAE = \frac{1}{n}\sum|e_t| = \frac{1}{5}(5+2+7+5+5) = 4.8$

(b) $MSE = \frac{1}{n}\sum e_t^2 = \frac{1}{5}(25+4+49+25+25) = 25.6$

(c) $RMSE = \sqrt{MSE} = \sqrt{25.6} = 5.06$

(d) $MAPE = \frac{1}{n}\sum\left|\frac{e_t}{y_t}\right| = \frac{1}{5}\left[\frac{5}{100} + \frac{2}{110} + \frac{7}{105} + \frac{5}{115} + \frac{1}{120}\right] = 4.40\%$

(e) RMSE is more sensitive to outliers because large errors makes them dominate the metric. MSE grows quadratically while MAE grows linearly.

Prob4 (a). $y_{10} = \varepsilon_{10} + \theta\varepsilon_9 = 2 - 0.6 = 1.4$

(b) $\hat{y}_{11|10} = E[y_{11}|\mathcal{F}_{10}] = E[\varepsilon_{11} + \theta\varepsilon_{10}|\mathcal{F}_{10}]$

since $E[\varepsilon_{11}] = 0$, knowing $\varepsilon_{10} = 2$, $E[\varepsilon_{10}|\mathcal{F}_{10}] = \varepsilon_{10}$

$\hat{y}_{11|10} = 0 + \theta\varepsilon_{10} = 0.6(2) = 1.2$

(c) $\hat{y}_{12|10} = E[\varepsilon_{12} + \theta\varepsilon_{11} | \mathcal{F}_{10}]$

$\varepsilon_{12}, \varepsilon_{11}$ are unknown at time 10.

$\therefore \hat{y}_{12|10} = 0$

(d) The MA(1) has short memory through finite shock propagation. Since MA(1) is $y_t = \varepsilon_t + \theta\varepsilon_{t-1}$, it contains No information about errors older than $t-1$, and only depends on the current error and previous error (one lag), i.e. dependence lasts exactly one period

(e) $\rho(1) = \theta/(1+\theta^2) = 0.4412$

Part II Github link.

https://github.com/YuqiaoLydia/ECON5140HM.git